

ELECT-SENTINEL OSINT

A Theoretical & Practical Framework for Real-Time Global Election Disinformation Monitoring, Dynamic Narrative Clustering, and Public Confusion Mitigation

Topic: Topic 1 — Election Disinformation & Public Confusion OSINT

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GitHub Repository:

<https://github.com/being-cheema/elect-sentinel-osint>

Roll Number: 23011103011

Operational Status: 100% Real Live Digital Streams (All Countries)

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1. Theoretical Foundations & Problem Statement

1.1 Information Disorder Taxonomy (Wardle & Derakhshan Framework): During critical election cycles, public discourse experiences heightened vulnerability to weaponized information. This research categorizes election-period confusion across three distinct conceptual dimensions: (1) *Disinformation* (knowingly fabricated content disseminated with malicious intent to suppress votes or alter outcomes), (2) *Misinformation* (false information shared inadvertently by confused citizens without malicious intent), and (3) *Malinformation* (genuine information taken out of context, selectively leaked, or distorted to inflict harm).

1.2 Cognitive Vulnerabilities & Epistemic Uncertainty: Misleading claims thrive during elections due to high affective polarization and epistemic uncertainty. When official counting protocols, ballot verification rules, or polling hours are ambiguous, unverified rumors spread rapidly through online echo chambers. The objective of **ELECT-SENTINEL OSINT** is to construct an automated intelligence system capable of monitoring multi-channel digital communications globally, identifying emerging confusion vectors before they peak, and equipping election oversight teams with rapid rebuttal intelligence.

2. Mathematical Formulations & Threat Modeling

The platform implements a multi-parameter mathematical model to evaluate threat severity, viral velocity, and narrative clustering similarity across digital posts:

$$S_{\text{confusion}} = [\alpha \cdot S_{\text{category}} + \beta \cdot S_{\text{urgency}} + \gamma \cdot S_{\text{uncertainty}} + \delta \cdot P_{\text{bot}}] \times \Omega_{\text{platform}}$$

Where:

- **S_{category} ($\alpha = 0.60$):** Semantic threat weight assigned based on matched disinformation taxonomies (Voter Suppression, Machine Tampering, Deepfakes, Official Impersonation, Premature Results, Intimidation).
- **S_{urgency} ($\beta = 0.20$):** Outrage and emotional urgency marker intensity calculated from linguistic triggers (e.g., 'URGENT SPREAD NOW', excessive capitalization, panic punctuation).
- **$S_{\text{uncertainty}}$ ($\gamma = 0.20$):** Epistemic rumor index derived from unverified hearsay qualifiers ('insider reveals', 'they don't want you to know', 'allegedly').
- **P_{bot} ($\delta = 0.15$):** Inauthenticity probability score derived from account age ratio, high posting burst rate, and copy-paste repetition patterns.
- **Ω_{platform} :** Platform risk coefficient ($\Omega \in [0.65, 1.25]$) weighted by source channel anonymity and moderation policies.

2.2 Dynamic Narrative Clustering & Lifecycle State Machine:

Individual posts are vectorized using n-gram Term Frequency-Inverse Document Frequency (TF-IDF):

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \log(|D| / |\{d \in D : t \in d\}|)$$

Incoming vectors are matched against active narrative centroids using Cosine Similarity: $\text{Sim}(v_p, v_N) = (v_p \cdot v_N) / (||v_p|| ||v_N||)$. If similarity exceeds the dynamic threshold ($\theta \geq 0.28$), the signal is assigned to the narrative cluster; otherwise, a new storyline centroid is spawned. Narrative lifecycle transitions through a discrete state machine: **Emerging** → **Accelerating** → **Critical Peak** → **Debunked** → **Dormant**.

3. Multi-Source Global Live Ingestion Engine

The system operates with **zero mock data**, pulling exclusively from 25+ real-time live digital feeds across the globe via a multi-threaded parallel architecture (ThreadPoolExecutor with 12 concurrent workers):

Category	Source Wires & Digital Endpoints	Geographic Scope & Purpose
Fact Checkers & Monitors	EUvsDisinfo (EU Strategic Disinformation DB), PolitiFact, Truth-O-Meter, FactCheck.org, FullFact UK, Google News Global Disinformation & Deepfakes Monitor	International claim refutation, synthetic media detection, and verified debunk repositories.
Global Broadcasters	BBC World News, Euronews International, The Guardian World, Al Jazeera English, Deutsche Welle (DW), France24 International	Worldwide continuous news wire monitoring in English, French, German, and Spanish.
Asia-Pacific & South Asia	The Hindu, Times of India, NDTV India, Google News Australia (Australian Electoral Commission & Federal Politics)	India, South Asia, and Commonwealth electoral monitoring.
Americas & Africa	Google News Canada, Google News US, Daily Maverick (South Africa), AllAfrica (40+ nations), MercoPress (Latin America / Mercosur)	North America, Pan-Africa, and South American electoral coverage.
Decentralized Social	Mastodon Public Federation Live Firehose (15+ international tags: #election, #voting, #democracy, #wahl, #elecciones, #eleicoes, #disinformation)	Real-time live social chatter across thousands of federated instance nodes.

4. Global Geocoding & Coordinated Inauthentic Behavior (CIB)

4.1 Global Geographic Entity Resolution: Content from any country is processed through a multi-national entity dictionary mapping over 60+ countries and capital jurisdictions (Americas, Europe, Asia-Pacific, Africa, Middle East, Oceania) to exact latitude and longitude coordinates. Resolved coordinates are dynamically rendered on the interactive Leaflet.js Dark Matter Map.

4.2 NetworkX Graph Modeling & CIB Swarm Detection: The system builds a multi-relational graph $G = (V, E)$ where nodes represent Accounts, Narrative Storylines, Hashtags, and Digital Domains. Edges capture posting relationships, copy-paste text similarity (Jaccard Index ≥ 0.65 within temporal windows), and cross-platform amplification. High-centrality hub accounts and coordinated astroturf clusters are highlighted automatically in real-time.

5. The 8 Analyst Workstations

- 1. Threat Radar:** Real-time DEFCON threat gauge (Critical/Elevated/Nominal), KPI dials, Category Threat Distribution Chart, Dissemination Bar Chart, and live stream ticker with audio alerts.
- 2. Narrative Intelligence:** Visual clustering hub tracking narrative lifecycles, viral velocity (posts/hr), cross-platform dispersion, and drill-down investigative links.
- 3. Analyst Triage Queue:** Multi-parameter filterable queue (Category, Platform, Priority P0-P3, Status, Min Confusion slider) with deep forensic inspector modal featuring SHA-256 cryptographic evidence hashes and triage status escalation.
- 4. Actor & Propagation Graph:** Interactive HTML5 Canvas physics network with spring-repulsion simulation, zoom/pan controls, draggable nodes, and CIB bot swarm community identification.
- 5. Global Geospatial Map:** Interactive dark-mode Leaflet.js map with localized threat circles colored by confusion index across 60+ countries worldwide.
- 6. Live OSINT Scanner:** Ad-hoc forensic investigation tool allowing analysts to paste any live URL, post, or claim from any country for instant threat evaluation and geocoding.
- 7. Ground Truth KB:** Repository of verified statutory election rules and voting security mandates with one-click rapid counter-messaging fact-check templates.

8. Intelligence Dossiers: Case Management workbench synthesizing formal OSINT Threat Intelligence Briefings exportable to Markdown, HTML, and printable format.

6. Empirical Verification, Testing & Benchmarks

The platform includes an automated Python test suite (tests/test_osint_engine.py) verifying NLP classification accuracy, statutory contradiction detection, dynamic clustering, and graph generation:

Test Suite Module	Evaluation Scope	Test Result
test_database_initialization	SQLite schema creation, ground truth seeding, and indices	PASSED (100%)
test_global_location_resolution	Entity extraction & geocoding across multi-continent text	PASSED (100%)
test_nlp_voter_suppression_detection	Confusion scoring & category classification for suppression	PASSED (100%)
test_nlp_machine_rigging_detection	Equipment tampering & election integrity conspiracy detection	PASSED (100%)
test_ground_truth_contradiction	Statutory rule contradiction checking & debunk generation	PASSED (100%)
test_narrative_clustering	TF-IDF semantic similarity grouping into active storylines	PASSED (100%)
test_network_graph_generation	NetworkX graph topology, edge weights, and CIB metrics	PASSED (100%)
test_case_and_report_generation	Formal Intelligence Briefing synthesis & Markdown formatting	PASSED (100%)

7. Evaluator Launch Guide & Execution Commands

To inspect and evaluate the project from the public GitHub repository:

```
# 1. Clone the public GitHub repository
git clone https://github.com/being-cheema/elect-sentinel-osint.git
cd elect-sentinel-osint

# 2. Start the application (creates venv, installs packages, starts live server)
chmod +x run.sh
./run.sh

# 3. Access the Live Cyber Operations Center in your web browser
http://localhost:8000

# 4. Run automated test suite
./venv/bin/python -m unittest tests/test_osint_engine.py
```

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GitHub Repository: <https://github.com/being-cheema/elect-sentinel-osint>
Submitted for Academic & Technical Evaluation. System is fully operational and continuously ingesting live global feeds.